

Trainee Program

Deshmukh Technologies Trainee Program

A 90-day full-stack software development and professional development program that prepares trainees to build, test, debug, explain, document, and maintain production software.

PRIMARY OUTCOME

Junior Full-Stack Developer

BACKEND TRACKS

Java / Spring Boot or Python / FastAPI

DELIVERY

Git + GitHub + Docker + CI/CD + AWS

FRONTEND

React + TypeScript

DATABASE

MySQL / PostgreSQL

DURATION

90 Days · 12 Phases

Learn. Build. Solve. Review. Deploy.

Grow.

INTERNAL TALENT DEVELOPMENT
STANDARD

DOCUMENT

DTTP Program
Handbook

VERSION

1.5

CLASSIFICATION

Internal Training
Standard

OWNER

Deshmukh
Technologies**HANDBOOK** Contents**PART I — PROGRAM FOUNDATION****01** Program Purpose**02** Program Vision**03** Program Mission**04** Development Model**05** Technology Architecture**06** Trainee Journey**07** 90-Day Master Roadmap**08** Training Methodology**09** Daily Development Cycle**PART II — TECHNICAL CURRICULUM****10** Phase 1 — Orientation**11** Baseline Assessment**12** Git & GitHub**13** Phase 2 — Programming**14** Java Track**15** Spring Boot**16** Python Track**17** FastAPI**18** Frontend**19** Database**20** REST API**PART III — PROJECT & ENGINEERING****21** Major Training Project**22** Full-Stack Architecture**23** Authentication & Security**24** Testing**25** Debugging**26** Docker**27** CI/CD**28** AWS**29** Project Management**PART IV — ASSESSMENT & CAREER****30** Definition of Done**31** Mentor System**32** Daily Report**33** Weekly Review**34** Competency Model**35** Performance Status**36** Weekly Score**37** Final Assessment**38** Graduation Levels**39** Production Readiness Gate**40** Career Progression**41** DTTP Management System**42** Final DTTP Standard

PART I Program Foundation

01 Program Purpose

The Deshmukh Technologies Trainee Program (DTTP) is a structured professional development program designed to transform freshers, interns, graduates, and entry-level candidates into capable software engineers.

The program combines technical education, practical coding, full-stack development, project-based learning, engineering practices, mentorship, code review, testing, debugging, security, DevOps, professional development, and continuous assessment.

The program does not define success as completion of 90 days. Success means the trainee can demonstrate the ability to build, test, debug, explain, document, and maintain software.

WHO THE PROGRAM IS FOR

- Freshers and recent graduates
- Interns converting into a structured path
- Career-switchers with basic programming exposure
- Entry-level hires who need a company standard

COMPANION DOCUMENTS

- This handbook — 90-day path and graduation
- Self-Growth Program — ten-level knowledge map
- Corporate Profile — identity, values, and method

The program is a supervised professional environment with attendance, reviews, Git history, and a production-readiness gate.

02 Program Vision

Build skilled, responsible and production-ready software professionals who can understand business requirements, develop complete applications, solve problems independently and contribute effectively to real Deshmukh Technologies projects.

03 Program Mission

1. Establish strong programming fundamentals.
2. Develop modern full-stack development skills.
3. Teach professional software engineering practices.
4. Provide practical project experience.
5. Develop independent problem-solving ability.
6. Build communication and teamwork skills.
7. Introduce testing, security and deployment.
8. Measure capability continuously.
9. Provide a structured path from trainee to developer.
10. Create a repeatable internal talent-development system.

04 Three-Dimensional Development Model

Every trainee is evaluated across three dimensions. A trainee must develop all three.

A. Technical

- Programming
- Frontend
- Backend
- Database
- APIs
- Testing
- Security
- Cloud

B. Engineering

- Git
- Architecture
- Clean code
- Debugging
- Code review
- CI/CD
- Docker
- Documentation

C. Professional

- Communication
- Discipline
- Teamwork
- Ownership
- Time management
- Accountability
- Problem solving

05 Technology Architecture

COMMON FOUNDATION

PROGRAMMING

OOP

DATA STRUCTURES

ALGORITHMS

WEB

SQL

GIT / GITHUB

REST APIS

FRONTEND

HTML

CSS

JAVASCRIPT

TYPESCRIPT

REACT

ENGINEERING

TESTING

SECURITY

DOCKER

CI/CD

AWS

MONITORING

BACKEND — TRACK A

JAVA

SPRING BOOT

SPRING DATA JPA

HIBERNATE

BACKEND — TRACK B

PYTHON

FASTAPI

SQLALCHEMY

06 Trainee Journey

01 Application	02 Screening	03 Technical Assessment
04 Interview	05 Selection	06 Orientation
07 Baseline Assessment	08 Foundation	09 Frontend
10 Backend Specialization	11 Database & APIs	12 Full-Stack Project
13 Testing	14 Security	15 Docker
16 CI/CD	17 AWS	18 Final Project
19 Final Assessment	20 Production Readiness	21 Junior Full-Stack Developer

07 90-Day Master Roadmap

Phase	Days	Outcome
1	1–5	Professional and engineering foundation
2	6–15	Programming and problem solving
3	16–25	Web and frontend foundation
4	26–35	React + TypeScript
5	36–45	Java / Spring Boot or Python / FastAPI
6	46–55	Database + REST API
7	56–65	Full-stack project
8	66–72	Authentication + security
9	73–78	Testing + debugging
10	79–82	Docker
11	83–86	CI/CD + AWS
12	87–90	Final project + assessment

PHASE DELIVERABLES

Phase	The trainee should produce
1	Baseline profile, professional standards acknowledgment, first Git repo
2	Solved problem set with commits and explanations
3–4	Responsive React pages for login, dashboard, and employee list
5–6	Working REST API and SQL schema for employees and departments
7	End-to-end create/list employee flow
8	Authenticated roles and protected routes
9	Automated tests and a written debug note for one real bug
10–11	Docker Compose runbook and a CI workflow
12	Final demo, documentation, and production-readiness evidence

WEEK-BY-WEEK OPERATING PLAN

Use this when a mentor asks “what should this trainee be doing right now?”

Days	Trainee work	Mentor looks for
1–5	Read the three official documents. Complete baseline. Open the first repo.	Can summarize how the company works. First daily report exists.
6–10	Language drills: types, functions, collections, errors.	Commits show small problems with input and output.
11–15	Beginner then intermediate problems. One oral walkthrough.	Can dry-run a loop and name the failing case.
16–25	HTML/CSS then JavaScript form, DOM table, fetch.	No React yet. Keyboard use. Fetch error state.
26–35	React layout, routes, login, protected employees, mocked service.	API calls are not inside JSX.
36–45	Backend folders, health, login, employee create/list.	400/401/409 are distinct. One service test.
46–55	Schema, joins, attendance, leave apply.	Join written without the ORM.
56–65	Wire React to the real API. Dashboard, profile, departments.	Network tab matches the contract.
66–72	JWT, roles, CORS, injection notes.	EMPLOYEE cannot create or approve.
73–78	Leave tests, one written bug, one measured query.	Failing test first. Two timings recorded.
79–86	Compose, Actions, AWS service-purpose notes.	CI fails on a broken test. Health endpoint works.
87–90	Stories, README, demo, oral review, readiness gate.	Trainee explains without reading slides.

08 Training Methodology

Recommended allocation of trainee time:

Theory		20%
Practical		30%
Project		40%
Review & Assessment		10%

The trainee should spend more time building software than listening to someone explain software.

09 Daily Development Cycle

LEARN

UNDERSTAND

PRACTICE

IMPLEMENT

TEST

DEBUG

COMMIT

PULL REQUEST

CODE REVIEW

IMPROVE

This should become the trainee's normal development habit.

PART II Technical Curriculum

10 Phase I — Orientation

Days 1–5. Orientation establishes company context and professional standards before technical training begins.

COMPANY

- Deshmukh Technologies
- Vision and mission
- Products and services
- Projects and teams
- Development methodology
- Customer expectations

PROFESSIONAL STANDARDS

- Attendance and working hours
- Communication and meetings
- Workplace conduct
- Confidentiality
- Information security
- Responsibility

ORIENTATION READING

Before Day 6, the trainee reads and can summarize the Corporate Master Profile, this handbook (purpose, daily cycle, Definition of Done), the Self-Growth Program, and the **Trainee Software Setup & Environment Verification** guide. Day 1 environment work follows that guide. Project work does not start until the mentor marks the environment Pass or Conditional.

Day	Focus	Written output
1	Company, values, and software environment sign-off	Setup evidence pack + one-page company summary
2	Professional standards and security	Signed acknowledgment + first daily report
3	Baseline assessment	Baseline Skill Profile
4	GitHub access and first repository	Empty project README and first commit
5	Mentor meeting	Week-1 plan with one measurable objective

11 Baseline Assessment

Before technical training begins, assess the trainee and record a **Baseline Skill Profile**.

PROGRAMMING

- Logic
- Variables
- Conditions
- Loops
- Functions

TECHNICAL

- Java
- Python
- JavaScript
- SQL
- Git

PROBLEM SOLVING

- Logical reasoning
- Coding problem

COMMUNICATION

- Self-introduction
- Technical explanation
- Problem explanation

HOW TO RECORD THE BASELINE

Score each area L0–L5. Write one sentence of evidence, not a slogan. The baseline is a starting point, not a grade.

Area	Level	Evidence
JavaScript	L2	Built a form and fetch demo with mentor help
SQL	L1	Can explain SELECT/WHERE, has not written a join
Git	L1	Has committed locally, has not opened a pull request

12 Git & GitHub

Git is mandatory. Trainees must learn repository, clone, commit, branch, push, pull, merge, rebase, stash, revert, pull request, code review, and conflict resolution.

STANDARD WORKFLOW

TASK	BRANCH	CODE	TEST	COMMIT	PUSH	PULL REQUEST	REVIEW	FIX	APPROVE	MERGE
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NAMING

```
Branch  dttp/<initials>/leave-approve-api
Commit  feat(leave): add approve endpoint and role check
```

Useful prefixes: feat, fix, test, docs, chore. One purpose per commit.

PULL REQUEST BODY

1. **What** changed
2. **Why** it changed
3. **How** it was tested
4. **What** a reviewer should look at first
5. **Risks** or follow-up work

A pull request with no test note is not ready for review.

```

### What
Add leave approve API for HR and ADMIN.

### Why
HR cannot currently decide a pending request.

### Test
- unit: employee cannot approve
- API: 401 / 403 / 409
- manual: HR approves one pending leave

### Review first
LeaveService.approve() role check

### Risk
Already-decided leaves must stay unchanged

```

REVIEW COMMENTS THAT HELP

Weak comment	Useful comment
“This is wrong.”	“Duplicate email should be 409 from the service, not 500 from the controller.”
“Clean this up.”	“Move <code>fetch('/api/employees')</code> into <code>employeeService.ts</code> .”
“Add tests.”	“Add <code>employee_cannot_approve</code> before merging.”
“Looks good.”	“Approve path is clear. Please add the already-decided 409 case.”

13 Phase 2 — Programming

Days 6–15.

[Variables](#)
[Data types](#)
[Operators](#)
[Conditions](#)
[Loops](#)
[Functions](#)
[Collections](#)
[Strings](#)
[Exceptions](#)
[OOP](#)

[Data structures](#)
[Algorithms](#)

BEGINNER EXERCISES

- Reverse string
- Palindrome
- Fibonacci
- Factorial
- Prime
- Largest element

INTERMEDIATE EXERCISES

- Two Sum
- Anagram
- Missing number
- Duplicate detection
- Frequency counting
- Sorting, searching, LinkedList

Every problem commit includes the prompt, one example input, the output, and the failing case. “Solved Two Sum” with no example is not accepted.

Problem	Example	Expected
Reverse string	DTTP	PTTD
Palindrome	level	true
Two Sum	[2, 7, 11, 15], target 9	[0, 1]
Duplicate detection	[1, 3, 3, 7]	3
Frequency count	leave, leave, sick	leave=2, sick=1

14 Java Track

JAVA FUNDAMENTALS

- OOP, classes, interfaces
- Inheritance, polymorphism, encapsulation
- Exceptions
- Collections
- Generics

MODERN JAVA

- Lambda
- Streams
- Optional
- Functional interfaces
- Date/time API

Expected evidence: an `Employee` class with private fields; a collection or stream that filters by department; `Optional` instead of `null`; a spoken difference between an interface and a class.

15 Spring Boot

CONTROLLER

SERVICE

REPOSITORY

JPA / HIBERNATE

DATABASE

Spring Boot

REST

Dependency Injection

DTO

Entity

Validation

Exception handling

Logging

JPA

Hibernate

Spring Security

JWT

Testing

```

EmployeeController.create(dto)
→ EmployeeService.create(dto)
    validate name/email
    reject duplicate email
    load department or fail
→ EmployeeRepository.save(entity)
→ 201 + EmployeeResponse

```

Layer	Owns	Must not own
Controller	HTTP status, DTO in/out	Unique-email rule, SQL
Service	Business rules, roles	Servlet / request objects
Repository	Persistence	Status codes
Entity	Table mapping	Password or JWT logic

16 Python Track

Variables	Lists	Tuples	Sets	Dictionaries	Functions	Modules	Packages	Exceptions	File handling
OOP	Type hints	Generators	Decorators	Async					

Expected evidence: a typed function with Pytest; grouping records with a dictionary; a Pydantic model or dataclass for an employee; one sentence each for list, tuple, and set.

17 FastAPI

ROUTER	SERVICE	REPOSITORY	SQLALCHEMY	DATABASE					
Routes	Request models	Response models	Pydantic	Validation	Dependency injection	Middleware			
Exceptions	Authentication	Async	API documentation						

```
employees.router.create(payload)
→ EmployeeService.create(payload)
    validate with Pydantic
    reject duplicate email
    load department or fail
→ EmployeeRepository.save(model)
→ 201 + EmployeeOut
```

Java and Python use different class names and the same rules: validation, unique email, role check, and one error JSON shape.

18 Frontend — React + TypeScript

Trainees learn components, props, state, hooks, forms, validation, routing, protected routes, API integration, error handling, loading states, and responsive design.

```
src/  
├ components/  
├ pages/  
├ layouts/  
├ services/  
├ hooks/  
├ types/  
├ utils/  
├ routes/  
└ assets/
```

Data	Lives in	Why
Login form fields	LoginPage local state	Only that page edits them
Auth token / current user	Auth context	Many routes need it
Employee list	Page state or hook	Fetches for that screen
API calls	services/*.ts	JSX must not contain URLs

Required screens by Day 35: /login, /dashboard, /employees (list + create), shared layout, empty/loading/error states.

SCREEN INVENTORY

Route	Role	Fields / actions	Empty / error copy
/login	Public	Email, password, Submit	“Email or password is wrong.”
/dashboard	All signed-in	Employees, present today, pending leave	“No data yet. Add employees first.”
/employees	ADMIN, HR	Name, email, department, joining date, role, Add	“No employees match this filter.”
/employees/new	ADMIN, HR	Name, email, department, joining date, role	Inline field errors
/employees/:id	ADMIN, HR; EMPLOYEE own	Same fields; EMPLOYEE cannot change role	“Employee not found.”
/departments	ADMIN, HR	Name, employee count, Add	“No departments yet.”
/attendance	ADMIN, HR; EMPLOYEE own	Date, employee, Present/Absent/Leave	“No attendance for this date.”
/leaves	All	Dates, type, status; Approve/Reject for HR/ADMIN	“No leave requests.”
/profile	All	Own name, email, department, password change	Validation on email and password

Shared layout: product name, signed-in user, role badge, logout. Missing token redirects to /login.

19 Database

Trainees learn SQL, tables, keys, constraints, relationships, joins, indexes, transactions, normalization, and query optimization.

ORM: Java uses JPA / Hibernate. Python uses SQLAlchemy.

ORM knowledge must not replace SQL knowledge.

```
departments(id, name UNIQUE)
employees(id, name, email UNIQUE, department_id FK, joining_date, role, password_hash)
attendance(id, employee_id FK, work_date, status, UNIQUE(employee_id, work_date))
leaves(id, employee_id FK, start_date, end_date, type, status)
```

Required by-hand queries: employees with department names (INNER JOIN); leave count by department (LEFT JOIN + GROUP BY); a transaction that approves leave and writes related attendance or rolls both back.

```
BEGIN;  
UPDATE leaves SET status = 'APPROVED' WHERE id = 18 AND status = 'PENDING';  
-- if row count is 0, ROLLBACK and return 409  
INSERT INTO attendance (employee_id, work_date, status)  
VALUES (41, '2026-08-20', 'LEAVE');  
COMMIT;
```

SEED DATA

Table	Rows
departments	Engineering, Human Resources, Finance
employees	Asha Patil (HR), Kiran Shah (ADMIN), Rohan Deshmukh (EMPLOYEE, Engineering)
attendance	Rohan Present on 2026-08-18 and 2026-08-19
leaves	Rohan 2026-08-20 to 2026-08-21, CASUAL, PENDING

Passwords are hashed. These values are examples, not production credentials.

20 REST API

Both backend tracks implement the same business API so Java and Python trainees learn the same API design principles.

Method	Endpoint	Purpose
POST	/api/auth/login	Authenticate user
GET	/api/employees	List employees
POST	/api/employees	Create employee
GET	/api/employees/{id}	Get employee
PUT	/api/employees/{id}	Update employee
DELETE	/api/employees/{id}	Delete employee
GET	/api/departments	List departments
POST	/api/departments	Create department
GET	/api/attendance	List attendance
POST	/api/attendance	Record attendance
GET	/api/leaves	List leaves
POST	/api/leaves	Apply for leave
PUT	/api/leaves/{id}/approve	Approve leave

ENDPOINT NOTE

Document every endpoint the trainee implements. Keep the notes in `docs/api.md` or OpenAPI. Do not leave the contract only in a mentor's memory.

Field	Example
Purpose	Create an employee
Roles	ADMIN, HR
Request	{ "name", "email", "departmentId", "joiningDate" }
Success	201 + employee JSON
Errors	400 validation, 401 missing token, 403 wrong role, 409 duplicate email

```
{
  "error": "VALIDATION_FAILED",
  "message": "Email is required",
  "field": "email"
}
```

REMAINING CONTRACTS THE TRAINEE MUST WRITE

Endpoint	Rule
GET /api/employees	200 array. Empty is [], not 404. Filter ?departmentId=3.
PUT /api/employees/{id}	Same roles as create. Unknown id 404. EMPLOYEE may change own name only.
DELETE /api/employees/{id}	ADMIN only, 204. Pending leave blocks delete with 409.
GET/POST /api/departments	Create requires unique name. Duplicate 409.
GET/POST /api/attendance	Body employeeId, workDate, status. Duplicate date 409.
POST /api/leaves	Dates and type. Employee id from token. Overlap 409. Starts PENDING.
GET /api/health	No auth. { "status": "ok" } for Compose and CI.

PART III Project & Engineering

21 Major Training Project

Employee Management System

Roles: ADMIN · HR · EMPLOYEE

Authentication	Dashboard	Employee Management	Department Management	Attendance	Leave	Reports
Profile						

Module	Minimum complete behavior
Authentication	Login, logout, invalid-credential handling, token stored safely
Dashboard	Role-aware summary of employees, attendance, and pending leave
Employees	Create, read, update, list, and validate required fields
Departments	Create and list; employee belongs to a department
Attendance	Record and list attendance for a date range
Leave	Apply, list own requests, approve/reject by HR or Admin
Reports	At least one SQL-backed report, such as leave count by department
Profile	View and update the signed-in user's own details

A module is not complete until it meets the Definition of Done, including tests, review, and documentation.

PROJECT README

The Employee Management repository must contain a README that another trainee can follow without asking the author.

1. What this system does
2. Roles (ADMIN / HR / EMPLOYEE)
3. How to run locally
4. How to run tests
5. Environment variables (names only, never secret values)
6. API summary or link to `docs/api.md`
7. Known limits

A repository with only source files and no README is not a complete project.

LEAVE APPROVAL — FULL PATH

1. EMPLOYEE submits `POST /api/leaves`. Status becomes `PENDING`.
2. HR opens the pending list. EMPLOYEE does not see the approve action.
3. HR calls `PUT /api/leaves/{id}/approve` with `{ "decision": "APPROVED" }`.
4. Service checks: caller is HR or ADMIN; leave exists; status is `PENDING`.
5. In one transaction the leave becomes `APPROVED`. Related attendance writes, if required, happen in the same transaction.
6. If the second write fails, both roll back and the leave stays `PENDING`.
7. A second approve returns `409`. Reject uses `"REJECTED"` and must not write attendance.

Actor	Apply	Approve	Can see
EMPLOYEE	Own	No	Own requests
HR	Yes	Yes	Pending / team
ADMIN	Yes	Yes	All

22 Full-Stack Architecture

JAVA IMPLEMENTATION

REACT	REST	SPRING BOOT	SERVICE	JPA / HIBERNATE	MYSQL / POSTGRESQL
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PYTHON IMPLEMENTATION

REACT	REST	FASTAPI	SERVICE	SQLALCHEMY	MYSQL / POSTGRESQL
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Both implementations must satisfy the same business requirements.

23 Authentication & Security

Authentication	Authorization	Password hashing	JWT	Roles	Permissions	CORS	Input validation
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SQL injection awareness

XSS fundamentals

Secrets management

Environment variables

HTTPS

Least privilege

Never commit passwords, API keys, tokens, private keys, or production credentials to GitHub.

LOGIN SEQUENCE

1. User posts { "email", "password" } to POST /api/auth/login.
2. Server verifies the password hash and issues a JWT with sub and role.
3. Frontend stores the token in memory or an httpOnly cookie — never in Git.
4. Later requests send Authorization: Bearer <token>.
5. Missing token → 401. Wrong role → 403.

Case	Status	Meaning
No token on /api/employees	401	Not authenticated
EMPLOYEE posts /api/employees	403	Authenticated, not allowed
HR posts a valid employee	201	Allowed and created
HR posts a duplicate email	409	Allowed, but conflicts

24 Testing

Backend

Java: JUnit, Mockito, Spring Boot tests

Python: Pytest, FastAPI testing, mocking

Frontend

- Component testing
- Form testing
- API-state testing
- Error-state testing

API

- Postman
- Positive and negative tests
- Authentication and authorization
- Edge cases

MINIMUM TEST CATALOG — LEAVE APPROVE

Test	Layer	Must prove
approve_pending_leave_as_hr	Service	Status becomes APPROVED
employee_cannot_approve	Service	Rule fails before save
approve_without_token	API	401
approve_as_employee	API	403
approve_already_decided	API	409
approve_unknown_id	API	404
create_employee_missing_email	API	400/422 with field email

A feature with only a happy-path test is not tested.

25 Debugging

BUG	REPRODUCE	READ ERROR	LOGS	NETWORK	API	BACKEND	DATABASE	ROOT CAUSE	FIX	RETEST
REGRESSION										

The trainee should learn to determine which layer is responsible rather than randomly changing code.

Symptom	Check first	Typical cause
Empty list, Network 401	Token header	Token missing, expired, or stored wrong
403 on create employee	Role in JWT	Logged in as EMPLOYEE
500 on duplicate email	Exception handler	Unique constraint not mapped to 409
Route blank after refresh	Protected route	Token read as empty string
API cannot reach DB in Compose	DB host name	Using localhost inside the API container
CI red, local green	Workflow commands	Tests not run the same way

26 Docker

FRONTEND CONTAINER	BACKEND CONTAINER	DATABASE CONTAINER
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Learn Dockerfile, image, container, network, volume, environment variables, and Docker Compose.

Service	Name	Port	Role
Frontend	ems - web	5173	React app
Backend	ems - api	8080	Java or Python API
Database	ems - db	3306 / 5432	MySQL or PostgreSQL

The API container talks to `ems - db`, not `localhost`. `docker compose up --build` must reach `GET /api/health`.

27 CI/CD



Job	Fails when
Lint	Formatting or unused imports break the standard
Unit tests	A service rule is wrong
Integration tests	API + database contract is wrong
Docker build	The image cannot be built from the branch
Health check	The app does not respond

The trainee must make CI fail on purpose once, then fix it, and keep both logs.

28 AWS



The trainee should understand the purpose of each service before using it.

Service	Purpose in DTTP language
IAM	Who is allowed to do what in AWS
EC2	A virtual machine that can run the app
S3	Object storage for files and build artifacts
RDS	Managed MySQL or PostgreSQL
CloudFront	CDN in front of static or public content
Route 53	DNS for the application hostname
Security Groups	Network allow/deny rules
CloudWatch	Logs and alarms

29 Project Management

Every trainee project should use a professional work breakdown.

Level	Example
Epic	Employee Management
Features	Authentication, Employees, Departments, Attendance, Leave, Reports
User Story	As an administrator, I want to create an employee so that the employee can be managed in the system.
Task	Create employee POST API.
Subtasks	Entity, DTO, Repository, Service, Controller, Validation, Test

This teaches trainees how professional teams convert requirements into engineering work.

PART IV Assessment & Career

30 Definition of Done

A feature is complete only when all of the following are true:

- | | |
|--|--|
| ☐ Requirement understood | ☐ Validation added |
| ☐ Acceptance criteria satisfied | ☐ Security checked |
| ☐ UI implemented | ☐ Tests written |
| ☐ API implemented | ☐ Tests passed |
| ☐ Database completed | ☐ Error cases checked |

- ☐ Git commit created
- ☐ Pull Request created
- ☐ Code review completed
- ☐ Review comments resolved
- ☐ Documentation updated
- ☐ Mentor approved

WHAT “DOCUMENTATION UPDATED” MEANS

Change	Update this
How to run or install	Project README
Endpoint, status, or payload	docs/api.md or OpenAPI
A non-obvious design choice	Short decision note
The day's work	Daily report with commit / PR link

“Documentation updated” is not a checkbox for later. If the notes are missing, the feature is not done.

31 Mentor System

Each trainee should have **1 Primary Mentor** and, where practical, **1 Technical Reviewer**.

MENTOR KPIS

Trainee progress	Code-review quality	Feedback quality	Skill-gap identification	Project completion
Production-readiness improvement				

This prevents mentorship from becoming informal or inconsistent.

WEEKLY MENTOR AGENDA

1. Git activity and pull requests
2. Assignment and project progress
3. One code-review example
4. One debugging or blocker discussion
5. Skill-gap and next-week plan
6. Status: Green / Amber / Red

INDIVIDUAL IMPROVEMENT PLAN

- The skill gap
- Expected evidence of improvement
- Practice work for the next 7 days
- Review date
- Mentor responsible

SAMPLE INDIVIDUAL IMPROVEMENT PLAN

```
Trainee: Rohan Deshmukh
Status: AMBER
Gap: Cannot separate 401 from 403; unique-email rule is still in the controller.
Evidence required by 22 Aug:
  1. Service rejects duplicate email; controller has no SQL.
  2. API tests: 401 / 403 / 409.
  3. Oral: explain both status codes without notes.
Practice: move the rule, write the failing test first, add API tests, oral review.
Mentor: Asha Patil
Review date: 22 Aug
```

If the evidence is missing on the review date, the status stays Amber or becomes Red. The plan is a written agreement.

32 Daily Report

Field	Question
Daily Objective	What was planned?
Completed	What was actually completed?
Learning	What was learned?
Problems	What went wrong?
Investigation	What was checked?
Resolution	How was it fixed?
Git Activity	Commits / PRs
Next Plan	Tomorrow's objectives

A report with only “worked on employees” is incomplete. Name the file, the endpoint, the error, or the commit.

SAMPLE DAILY REPORT

Field	Example
Daily Objective	Add validated POST /api/employees on the Java track
Completed	Endpoint returns 201; duplicate email returns 409
Learning	Unique constraint belongs in the service and the database
Problems	First attempt returned 500 on duplicate email
Investigation	Read Hibernate exception and mapped it in the exception handler
Resolution	Service throws ConflictException; handler returns 409 JSON
Git Activity	<code>feat(employees): reject duplicate email · PR #14</code>
Next Plan	Add GET list with department name join

33 Weekly Review

The mentor reviews technical progress, assignment completion, project progress, code quality, problem solving, communication, discipline, Git activity, blockers, and skill gaps.

WEEKLY REVIEW NOTE

The mentor writes a short note the trainee can act on. Verbal feedback that is not written down does not count as a weekly review.

```
Progress: employee create/list works with auth.
Quality: controller still contains the unique-email rule.
Gap: cannot explain 401 vs 403 without notes.
Next week: move the rule to the service; add 401/403 API tests.
Status: AMBER
```

34 Competency Model

Level	Definition
L0	Not exposed
L1	Understands
L2	Performs with guidance
L3	Performs independently — graduation target for core skills
L4	Can review / explain
L5	Can own

Advanced areas such as architecture, CI/CD, and AWS may initially target **L2**.

35 Performance Status

● **Green**

On track.

● **Amber**

Needs improvement.

● **Red**

At risk.

Amber and Red trainees receive an **Individual Improvement Plan**.

36 Weekly Score

Area	Weight
Technical Learning	15
Practical Work	20
Problem Solving	15
Project Contribution	20
Code Quality	10
Testing	5
Communication	5
Discipline	5
Documentation	5
Total	100

Score band	Meaning
85–100	Independent, reviewed, documented, and reusable work
70–84	Correct work with guidance and a few review cycles
55–69	Incomplete, fragile, or poorly explained work
Below 55	Missing fundamentals or professional standards

A high weekly score requires Git history, not only a verbal update.

WHAT THE BANDS LOOK LIKE ON CREATE EMPLOYEE

Band	What the mentor sees
85–100	Layered code, 409 mapped, tests, PR note, trainee explains the rule without slides
70–84	Feature works after two review rounds; tests cover the happy path only
55–69	Create works; duplicate email is 500; trainee cannot name the layer
Below 55	No validation, secrets in Git, or no pull request

37 Final Assessment

Category	Weight
Programming	15%
Frontend	10%
Backend	15%
Database / API	10%
Final Project	20%
Problem Solving	10%
Testing / Debugging	5%
Git / CI/CD	5%
Documentation	5%
Communication / Professionalism	5%
Total	100%

WHAT A DEVELOPER READY (85+) DEMO PROVES

1. Logs in as HR and as EMPLOYEE and shows the different menus.
2. Creates an employee and shows the 201 in the Network tab.
3. Repeats the same email and shows 409.
4. Applies leave as EMPLOYEE, approves as HR, retries and shows 409.
5. Opens the SQL for leave count by department.
6. Points to the service test that blocks EMPLOYEE approve.
7. Starts the stack with Compose and hits `/api/health`.
8. Names one next step and one thing they still cannot own.

A 70–84 trainee can do most of this with mentor prompts. A 55–69 trainee cannot complete leave approve or cannot explain a status code.

TEN-MINUTE DEMO SCRIPT

Minute	Show
0–1	Problem: HR cannot manage people or leave in one system
1–3	Login as HR. Dashboard counts. Employee list.
3–5	Create employee. Duplicate email. Network statuses.
5–7	EMPLOYEE apply leave. HR approve. Retry 409.
7–8	SQL report. One test name and what it proves.
8–9	Compose / health. What is not done yet.
9–10	Questions

38 Graduation Levels

85–100 • Developer Ready

Eligible for Junior Full-Stack Developer responsibilities.

70–84 • Junior Developer — Supervised

Can contribute with closer mentoring.

55–69 • Extended Training

Requires additional structured training.

Below 55 • Management Review

Formal decision required.

39 Production Readiness Gate

Before graduation, the trainee must demonstrate:

☐ Requirement understanding

☐ System design

☐ React development

☐ Backend development

☐ SQL

☐ REST APIs

☐ Authentication

☐ Authorization

☐ Testing

☐ Debugging

☐ Git

☐ Pull Requests

☐ Code review

☐ Docker

☐ CI/CD understanding

☐ Deployment understanding

☐ Documentation

☐ Technical communication

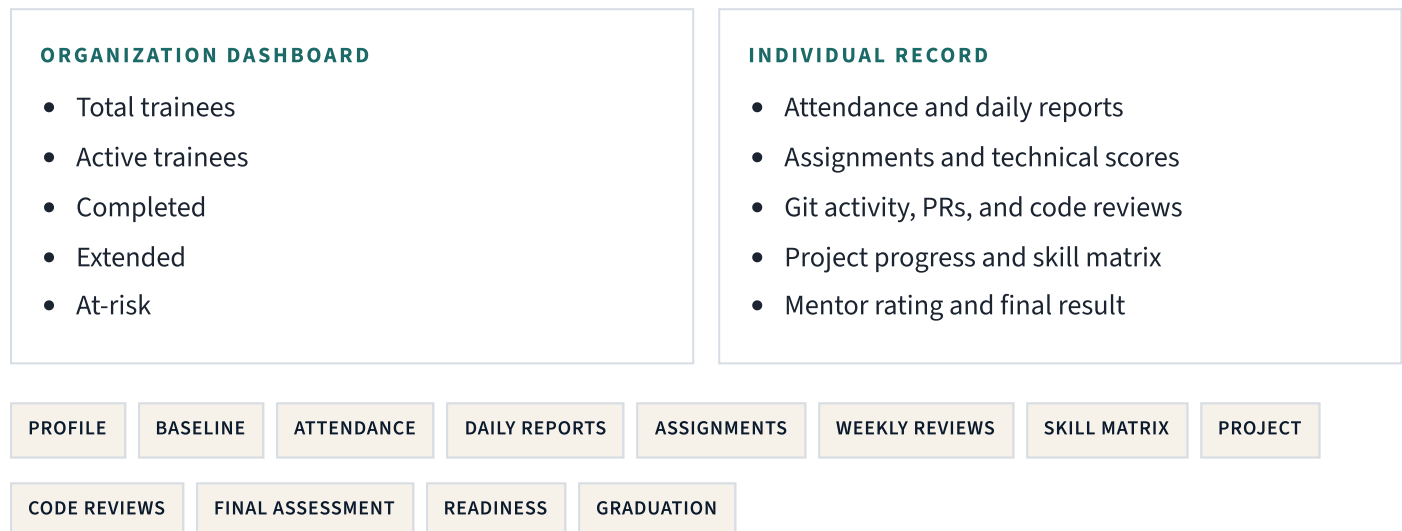
40 Career Progression



Promotion must be based on **demonstrated capability**, not simply completion of the 90-day period.

41 DTTP Management System

The long-term goal is an internal Trainee Management System.



42 Final DTTP Standard

The program should not ask: “How many technologies did the trainee complete?”

What can this trainee independently build, test, debug, explain, document and maintain?



DESHMUKH TECHNOLOGIES

DTP — Deshmukh Technologies Trainee Program

Learn. Build. Solve. Review. Deploy. Grow.

A disciplined, technically capable and production-
ready Junior Full-Stack Developer.